

Data 498D Senior Capstone

MWF 11:00AM-11:50AM

Classroom: Psych 204

Version as of 07.10.2026

Course Description

A culminating “capstone” experience that synthesizes concepts and skills for learning from data, as accumulated over the students’ experience in the Data Science major. We will introduce students to activities that will round out their undergraduate experience, including, concepts in reproducible research, ethics in statistical and data science practice, high performance computing, written and oral communication skills, working in teams, and professional development. Students will engage in projects and work to complete their academic portfolio.

Course Prerequisites: Introduction to Statistical Computing (DATA 375)

Course Co-requisites: Applied Linear Models (DATA 467) OR Principles of Machine Learning (CSC 480) OR Big Data in Molecular Biology and Biomedicine (MCB 447)

Instructor and Contact Information

Instructors:

Joseph Watkins, jwatkins@arizona.edu

Lianfen Qian, lqian01@arizona.edu

Office Hours: Posted in D2L Brightspace

D2L Brightspace Link: <https://d2l.arizona.edu/d2l/home/1780657>

Course Format and Teaching Methods

The course meets in person three times per week. It utilizes “flip classroom” format. Students complete their work through Brightspace. Access to all course materials such as readings, reading assignments, websites, or videos is provided on the Brightspace course site. Students may not modify content or re-use content for any purpose other than personal educational reasons. Students accessing unauthorized materials or using them in a manner inconsistent with UArizona values and educational policies are subject to suspension. All recordings and materials are the property of the faculty members.

Course Objectives: During this course, students will:

- Complement their statistical and data science skills with modern internet hosting services for software development and version controlling.
- Understand the obligations necessary for the ethical practice of data science
- Improve their oral and written communication skills.
- Work collaboratively on a substantial data intensive project.
- Review and improve past projects and add a substantial new project to their academic portfolio.
- Become more prepared for the next steps for a career in industry, government, or academia.

Expected Learning Outcomes: Upon completion of this course, students should be able to:

- Complement their statistical and data science skills with modern internet hosting services for software development and version controlling.
- Understand the obligations necessary for the ethical practice of data science
- Improve their oral and written communication skills.
- Work collaboratively on a substantial data intensive project.
- Review and improve past projects and add a substantial new project to their academic portfolio.
- Become more prepared for the next steps for a career in industry, government, or academia.

Absence and Class Participation Policy

Participating in the course and attending lectures and other course events are vital to the learning process. As

such, attendance is required at all lectures and discussion section meetings. Absences may affect a student's final course grade. If you anticipate being absent, are unexpectedly absent, or are unable to participate in class activities, please contact us as soon as possible. The UA's policy concerning Class Attendance and Participation is available at: <https://catalog.arizona.edu/policy/courses-credit/courses/class-attendance-participation>

- The UA policy regarding absences for any sincerely held religious belief, observance or practice will be accommodated where reasonable, <http://policy.arizona.edu/human-resources/religious-accommodation-policy>.
- Students who need to miss more than 1 week of classes (or the equivalent for accelerated courses) should provide documentation to the Dean of Students office (DOS-deanofstudents@arizona.edu). Documentation should be submitted prior to a planned absence. Documentation may be submitted during or after an unplanned absence.
- It is the student's responsibility to notify the instructor in advance of an absence related to religious observation or an activity for which a dean's excuse has been granted, and to arrange for how any missed work will be handled. See more information here: <https://policy.arizona.edu/student-life-and-education/attendance>.

Demonstrating Attendance and Participation

Students must demonstrate their attendance and participation in the course by attending the in-person class meetings and complete activities according to the due dates posted on D2L course site.

Students who have not completed the first week's assignments by due dates will be administratively dropped with deletion from their record. This means that it doesn't appear on the student transcript.

Makeup Policy for Students Who Register Late

Students who register after the first day of classes are responsible for notifying the instructors of the course by the beginning of the next class meeting after they have registered. Makeup work will be provided at the instructor's discretion.

Course Communications

Announcements and important course information may be sent out via official University email through D2L Brightspace. It is the student's responsibility to check messages and announcements regularly.

Required Texts or Readings

None required. Supplemental readings will be provided through D2L Brightspace throughout the course where necessary.

Required Materials

- Computer: For this class you will need daily access to a laptop or web-enabled device, and regular access to reliable internet signal. Our coursework will take place in D2L Brightspace and Gradescope. Note: Enrolled students can borrow technology from the UA Library on a first come, first served basis. See <https://new.library.arizona.edu/tech/borrow> for details.
- Gradescope: You will access Gradescope through the D2L Brightspace site. Please attempt to access the site as soon as possible to address any potential problems you may have. This is the platform where you will submit your written homework assignments each week.

Assignments: There are 3 components to your overall grade:

- Class Activities and Discussions (30% of course grade) – This class is largely a discussion-based class. Active participation in all discussions and in course activities is expected. These will be graded on participation and correctness.
- Portfolio (20% of course grade) – Part of becoming a professional includes creating and maintaining a professional portfolio. Throughout the semester you will work on creating and adding things to your portfolio. This will be submitted at the end of the semester for evaluation. Feedback opportunities will be provided throughout the

semester.

- Projects (50% of course grade): This course includes two projects that will be completed throughout the semester.
 - Project 1 (20% of course grade): This is an individual project to update and improve upon a project that you completed earlier in your academic career. The components to this project include
 - Reflection Essay (50% of the project grade) – A reflection on your initial work.
 - Poster (50% of the project grade) – Created based on the updates to the initial work.
 - Project 2 (30% of course grade): This project will be completed in groups of 2-3 students. The components to this project include
 - Proposal (5% of project grade)
 - Midterm Report (15% of project grade)
 - Final Report (30% of project grade)
 - Slides and Presentation (50% of project grade)

All grading disputes must be addressed within one week of an assignment being returned.

Final Examination

In lieu of a final exam, students will present their project 2 to the class in person. Everyone will speak a small portion of the presentations. Grading of the written portion of the projects will be completed using the rubric that is used to grade the midterm projects. Grading of the presentation will be completed using a rubric based on components of innovation, research, analysis and group work.

The University final exam schedule and rules relating to final examinations may be found at: <https://registrar.arizona.edu/finals>.

Grading Scale and Policies

Grades	Percent of Grade	You are Guaranteed a Grade of:
In Class Activities	30%	A if you earn at least 90%
Portfolio	20%	B if you earn at least 80%
Project 1	20%	C if you earn at least 70%
Project 2	30%	D if you earn at least 60%
Total	100 %	

Please note that neither exam scores nor final grades will be curved. No extra credit or bonus points are offered in this course.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with university policy, which is available at <https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>.

A student may withdraw from the course without a grade of W by September 6th, 2026, using UAccess. A student may withdraw with a grade of "W" through November 1st, 2026. It is suggested that students consult their academic advisor before withdrawal from any course.

Tentative Schedule: Topics/Activities

Week	Topics	Assignments Due
1	Introduction	Class Activities & Gradescope
2	Github	Class Activities & Gradescope
3	Written Communication	Class Activities & Gradescope
4	Written Communication	Class Activities & Gradescope
5	Professional Development	Class Activities & Gradescope
6	Data/Document Management	In Class Activities; Portfolio Check
7	Latex	Class Activities & Gradescope & Reflection Essay
8	Reproducible Research	Class Activities & Gradescope & Project 1 (Poster and Report)
9	Markdown	Class Activities & Gradescope
10	Ethics	Class Activities & Gradescope
11	Linux/ High Performance Computing Markdown	Class Activities & Gradescope
12	Working with Teams	Class Activities & Gradescope
13	Verbal Communication	Class Activities & Gradescope
14	Final Project	Class Activities & Gradescope
15	Final Project	Class Activities & Gradescope
16	Final Project	Project 2 Presentation & Paper; Final Portfolio

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself. See <http://policy.arizona.edu/education-and-student-affairs/threatening-behavior-students>.

Accessibility and Accommodations

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, <https://drc.arizona.edu/>) to establish reasonable accommodations.

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See: <https://deanofstudents.arizona.edu/policies/code-academic-integrity>.

The University Libraries have some excellent tips for avoiding plagiarism, available at <http://new.library.arizona.edu/research/citing/plagiarism>.

Selling class notes and/or other course materials to other students or to a third party for resale is not permitted without the instructor's express written consent. Violations of this and other course rules are subject to the Code of Academic Integrity and may result in course sanctions. Additionally, students who use D2L Brightspace or UA (University of Arizona) e-mail to sell or buy these copyrighted materials are subject to the Code of Conduct Violations for misuse of student e-mail addresses. This conduct may also constitute copyright infringement.

Generative AI

In this course all uses of generative artificial intelligence (AI)/large language model tools such as ChatGPT, Dall-e, Google Bard, Microsoft Bing, etc. may be considered a violation of the Code of Academic Integrity, specifically the prohibition against submitting work that is not your own. This applies to all assessments in the course, including written assignments, discussions and projects. This course policy is driven by the learning goals and desired learning outcomes for the course. Please ensure your use of AI is responsible and contributes to your understanding. We don't want to read AI generated responses, we much prefer to learn from you as an individual – you are much more interesting!

Nondiscrimination and Anti-harassment Policy

The University of Arizona is committed to creating and maintaining an environment free of discrimination. In support of this commitment, the University prohibits discrimination, including harassment and retaliation, based on a protected classification, including race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, or genetic information. For more information, including how to report a concern, please see <http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy>

Our classroom is a place where everyone is encouraged to express well-formed opinions and their reasons for those opinions. We also want to create a tolerant and open environment where such opinions can be expressed without resorting to bullying or discrimination against others.

Additional Resources for Students

UA Academic policies and procedures are available at <http://catalog.arizona.edu/policies>

Campus Health

<http://www.health.arizona.edu/>

Campus Health provides quality medical and mental health care services through virtual and in-person care.

Phone: 520-621-9202

Counseling and Psych Services (CAPS)

<https://health.arizona.edu/counseling-psych-services>

CAPS provides mental health care, including short-term counseling services.

Phone: 520-621-3334

The Dean of Students Office's Student Assistance Program

<https://deanofstudents.arizona.edu/support/student-assistance>

Student Assistance helps students manage crises, life traumas, and other barriers that impede success. The staff addresses the needs of students who experience issues related to social adjustment, academic challenges, psychological health, physical health, victimization, and relationship issues, through a variety

of interventions, referrals, and follow up services.

Email: DOS-deanofstudents@email.arizona.edu

Phone: 520-621-7057

Survivor Advocacy Program

<https://survivoradvocacy.arizona.edu/>

The Survivor Advocacy Program provides confidential support and advocacy services to student survivors of sexual and gender-based violence. The Program can also advise students about relevant non-UA resources available within the local community for support.

Email: survivoradvocacy@email.arizona.edu

Phone: 520-621-5767

Campus Pantry

Any student who has difficulty affording groceries or accessing sufficient food to eat every day, or who lacks a safe and stable place to live and believes this may affect their performance in the course, is urged to contact the Dean of Students for support. In addition, the University of Arizona Campus Pantry is open for students to receive supplemental groceries at no cost. Please see their website at: campuspantry.arizona.edu for open times.

Furthermore, please notify me if you are comfortable in doing so. This will let me provide any resources I may have.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): <https://cirt.arizona.edu/case-emergency/overview>

Confidentiality of Student Records

<http://www.registrar.arizona.edu/ferpa>

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.

University-wide Policies link

Links to the following UA policies are provided here, <http://catalog.arizona.edu/syllabus-policies>:

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy
- Subject to Change Statement